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THE PRESENT AND FUTURE

JACC STATE-OF-THE-ART REVIEW

"Food Is Medicine" Strategies for Nutrition Security and Cardiometabolic Health Equity

JACC State-of-the-Art Review

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ABSTRACT

"Food Is Medicine" (FIM) represents a spectrum of food-based interventions integrated into health care for patients with specific health conditions and often social needs. Programs include medically tailored meals, groceries, and produce prescriptions, with varying levels of nutrition and culinary education. Supportive advances include expanded care pathways and payment models, e-screening for food and nutrition security, and curricular and accreditation requirements for medical nutrition education. Evidence supports positive effects of FIM on food insecurity, diet quality, glucose control, hypertension, body weight, disease self-management, self-perceived physical and mental health, and cost-effectiveness or cost savings. However, most studies to date are quasiexperimental or pre/post interventions; larger randomized trials are ongoing. New national and local programs and policies are rapidly accelerating FIM within health care. Remaining research gaps require rigorous, iterative evaluation. Successful incorporation of FIM into health care will require multiparty partnerships to assess, optimize, and scale these promising treatments to advance health and health equity. (J Am Coll Cardiol 2024;83:843-864) © 2024 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Cardiovascular disease (CVD) is the leading cause of mortality in the United States and worldwide, with worsening trends across low-, middle-, and high-income countries.[1,2] Among modifiable metabolic and behavioral risks for CVD, poor diet has emerged as a top contributor, estimated to cause 45% of U.S. cardiometabolic deaths, 36% of global coronary deaths, 70% of new diabetes cases worldwide, and large burdens of disability and premature mortality.[3-5] In the United States, an increase in age-adjusted CVD mortality since 2011 marked the end of a 40-year decline,[1] owing to rising diet-related diabetes mellitus, adiposity, and related conditions including atherogenic dyslipidemia, hypertension, and inflammation.[6] Cardiometabolic risk is attributable to low intakes of protective dietary

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ABBREVIATIONS AND ACRONYMS

BMI = body mass index

CVD = cardiovascular disease

EHR = electronic health record

FIM = Food is Medicine

MNE = medical nutrition education

MTM = medically tailored meal

RDN = registered dietitian nutritionist

WIC = Special Supplemental Nutrition Program for Women, Infants, and Children

factors (eg, fruits, vegetables, whole grains, legumes, unsaturated fatty acids, fish), and high intakes of harmful components (eg, refined grains, processed meats, saturated fat, sodium, added sugars).[2-5,7] Worldwide, population-wide success at eating a healthy diet (measured by the validated Alternative Healthy Eating Index, Dietary Approaches to Stop Hypertension, and Mediterranean Diet scores) is poor, with typical average scores of 30 to 45 out of 100 (most healthy) across world regions.[8]

Cardiometabolic diseases and their lifestyle drivers, including low diet quality, also cause striking health disparities by income, geographic residence, race, and ethnicity.[1,2,8,9] These disparities are strongly associated with social determinants of health such as food insecurity—a household-level social or economic condition of limited access to sufficient food[10,11]—as well as lower education, income, housing opportunities, and health care access, which in turn relate to longstanding geographic inequities and ethnic disparities. In this way, suboptimal nutrition and food insecurity are important mediators of health disparities. Cardiometabolic diseases also generate substantial economic losses. Direct and indirect economic costs of CVD in the United States were estimated to be $405 billion annually in 2019,[1] with a projected doubling to $800 billion by 2030, and those of diabetes to be $327 billion in 2017,[12] a 26% increase in just 5 years. Suboptimal nutrition causes a meaningful proportion of these costs.[13-15] Evaluating U.S. medical costs and lost productivity due to poor nutrition, total costs were estimated to be $963 billion annually, and those due to food insecurity to be another $146 billion, a total of $1.1 trillion in yearly economic losses.[15]

The public health, equity, and economic impacts from poor diet have prompted calls for the health sector to play a greater role in addressing nutrition.[16,17] This increasing spotlight further aligns with health system transitions toward population health and health equity; new care and payment models that prioritize value-based care, disease management, and prevention; expanded insurance coverage for lower-income individuals; and growth in digital health information, food purchasing, and delivery systems. Clinical practice and population guidelines also emphasize nutrition for primary and secondary prevention of CVD[18] and as a cornerstone of cardiovascular health across the lifespan.[19,20] Yet historically, there have been few practical treatment options for clinicians to address their patients' diets, compounded by insufficient medical nutrition education (MNE), electronic nutritional screening or decision support, team-based care, or health policies, including incentive and reimbursement mechanisms to support such an infrastructure. Similarly, burdens of poor nutrition for patients are rarely driven by individual knowledge or choice alone, but influenced by complex social, environmental, cultural, agricultural, political, and economic determinants.[9] Accordingly, practical effective health sector interventions to improve nutrition, diet-related diseases, and related disparities have been relatively lacking. The unfortunate irony is that health systems have thus historically not addressed a top driver of poor health.

HIGHLIGHTS

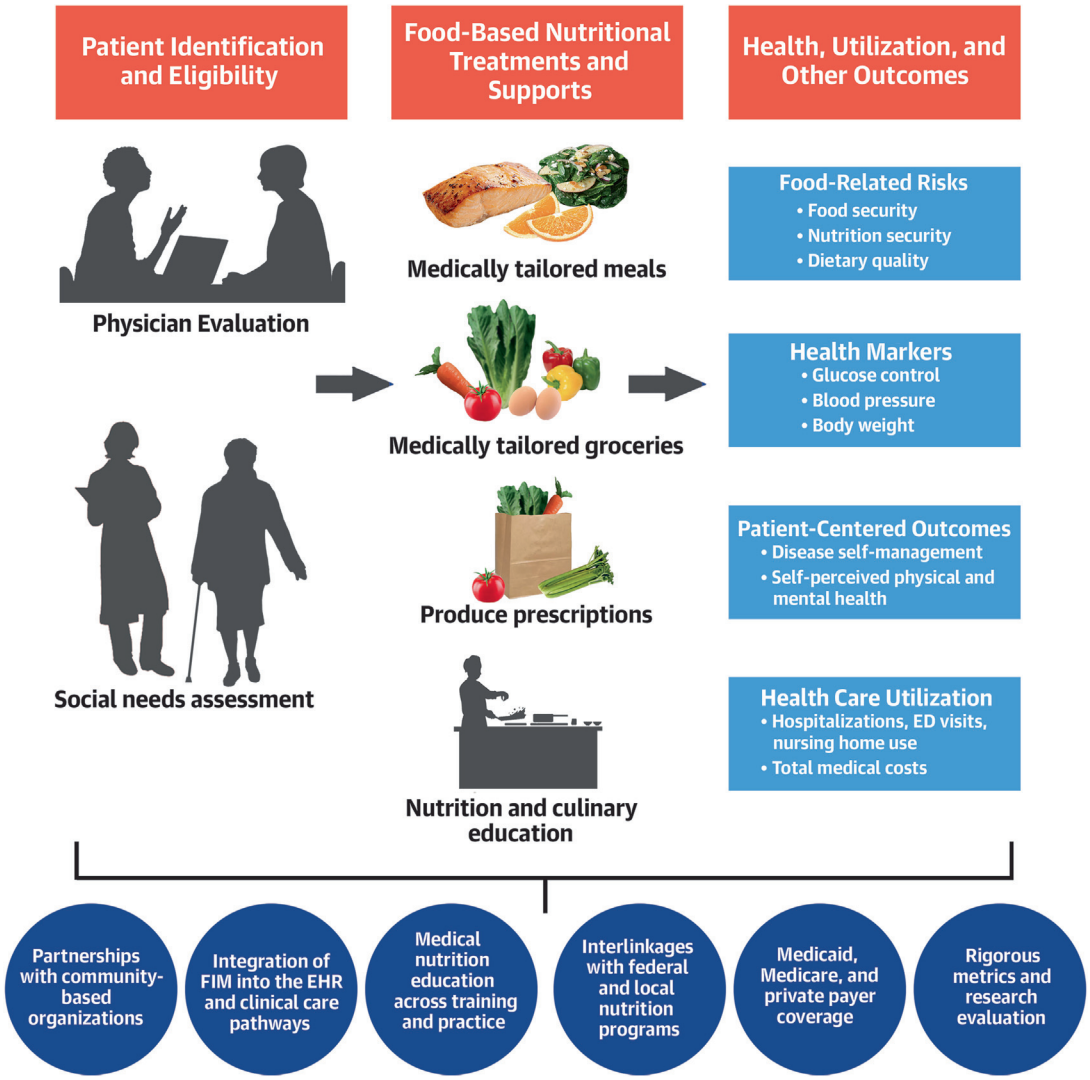
- "Food is Medicine" interventions integrate food-based treatments into health care systems.
- Preliminary evidence suggests significant positive impacts of these programs.
- Clinicians, health systems, and payers should support current and future directions of this strategy.

This is now changing across U.S. medical care and health policy with the advent of "Food Is Medicine" (FIM) initiatives. FIM can be defined as food-based nutritional interventions integrated within health systems to treat or prevent disease and advance health equity (**Central Illustration**).[21-23] These programs are often combined with nutrition and culinary education and are supported by new clinical care pathways and payment models, electronic health record (EHR) screening for food and nutrition security and diet quality, and greater MNE. In this accelerating landscape, it is important for cardiology care teams and other clinicians to understand this growing field and its potential for advancing health and health equity. This includes criteria for risk assessment and referral, and development of implementable frameworks to integrate FIM into health systems and clinical care.[24] This *JACC* State-of-the-Art Review assesses and discusses the current knowledge regarding FIM, including the: 1) advances in underlying science; 2) evidence for effectiveness; 3) intersections with food security, nutrition security, and health equity; and 4) corresponding new national, state, and private health policies and initiatives.

OVERVIEW OF FIM

FIM can be conceptualized as a spectrum of programs and services within or supporting health care systems

CENTRAL ILLUSTRATION A "Food is Medicine" Approach in Health Care



Mozaffarian D, et al. J Am Coll Cardiol. 2024;83(8):843-864.

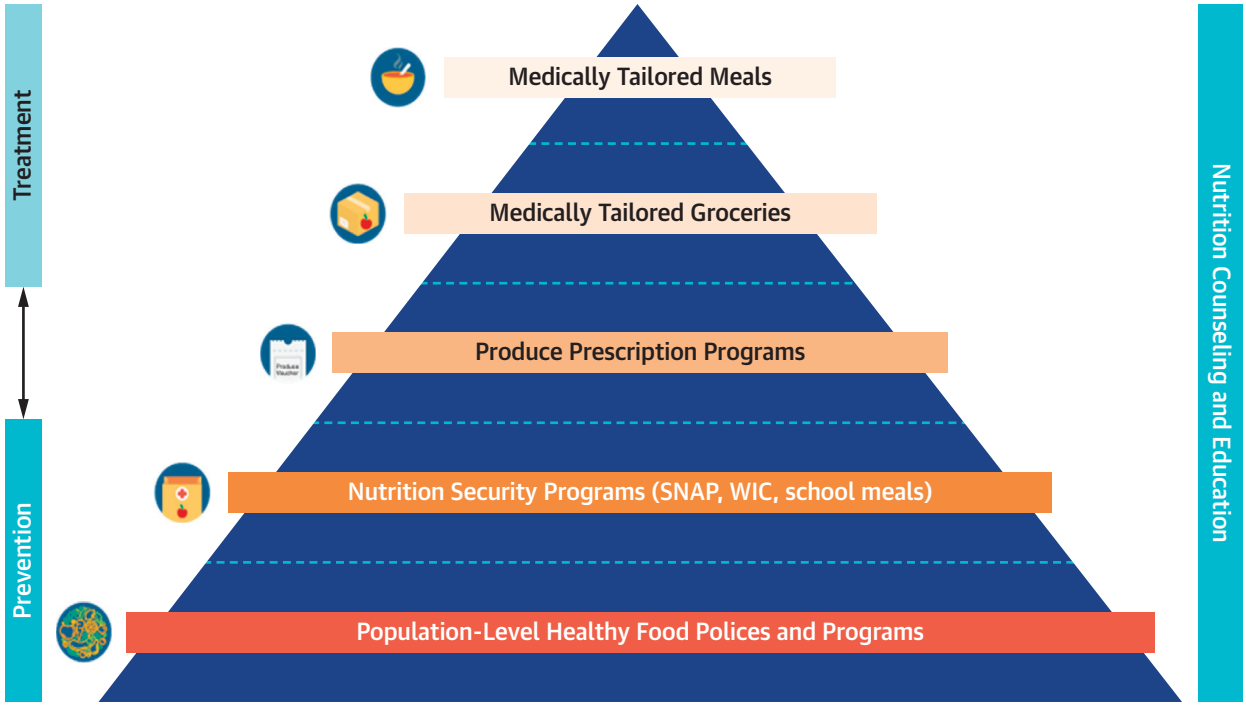
Food Is Medicine (FIM) interventions are food-based nutritional treatments, integrated into health care, to treat and advance health equity among patients with specific disease conditions and often social needs. These programs often include nutrition and culinary education and can benefit from supportive partnerships, e-screening and care pathways, medical nutrition education for clinicians, interlinkages with food assistance programs, and payer coverage. ED = emergency department; EHR - electronic health record.

(**Figure 1**).[21-23] Higher levels of the spectrum include more intensive interventions for sicker patient groups with complex medical conditions and high health care utilization. At progressively lower and broader levels, programs focus on larger and more general populations, ultimately founded on population-level policies that support healthier eating.

Several aspects of FIM are innovative. First, FIM interventions aim to address both poor nutrition and food insecurity.[25-27] This interrelationship has been conceived as "nutrition security": access, availability, and affordability of foods and beverages that promote well-being and prevent or treat disease.[28] This builds on the framework of food security to consider not only sufficient calories, but sufficient nourishment.

JACC VOL. 83, NO. 8, 2024
FEBRUARY 27, 2024:843-864

FIGURE 1 A Framework for Food Is Medicine Integrated Into Health Care System



Food Is Medicine can be conceptualized as a spectrum of programs, services, and other interventions, integrated within or supportive of health care systems, that recognize and respond to the critical link between nutrition and health. Lower levels of the spectrum focus on population-level prevention of disease through broad policies and programs. Shifting from prevention toward treatment, higher-level interventions are medical treatment programs that target patient groups with specific medical conditions as well as social needs. Nutrition counseling and education should be integrated across these strategies. Adapted with permission from Deuman et al.[59] SNAP = Supplemental Nutrition Assistance Program; WIC = Special Supplemental Nutrition Program for Women, Infants, and Children.

The relevance of nutrition security was highlighted during the COVID-19 response, which exacerbated both food insecurity and poor nutrition in vulnerable populations, and diet-related chronic conditions were major determinants of poor outcomes and disparities from COVID-19.[29]

Second, FIM interventions focus on increasing access, availability, affordability, and consumption of healthful foods. This is important because cardiometabolic risk is attributable to both insufficient intakes of beneficial foods and excessive intakes of harmful foods.[2-5,7] Yet many conventional nutrition policies to address cardiometabolic health have focused on isolated factors thought to be harmful, such as sodium, added sugar, total and saturated fat, dietary cholesterol, and, more recently, total calories. Such strategies often mirror the "tobacco playbook," derived from the 2005 Framework Convention on Tobacco Control, emphasizing negative messaging/disincentives such as front-of-pack labeling, menu labeling, warning labels, taxation, and voluntary or mandatory nutrient content limits. In contrast, few widespread nutritional policies have prioritized positive strategies to increase intakes of protective foods. FIM can help address this important gap in the clinical and policy armamentarium to improve cardiometabolic health.

Finally, FIM is grounded in health equity, focusing on the care and well-being of vulnerable populations and addressing some of the structural barriers to their nutrition and health. Often, innovations in clinical medicine may not improve, or can even worsen, health disparities by being less available and less prescribed to patients who need them most. However, FIM programs are designed to target and reach vulnerable populations with specific medical conditions who are experiencing food insecurity, financial strain, or other social challenges.[23] Thus, FIM interventions hold promise as a clinical innovation that can jointly improve health and address health equity.

MEDICALLY TAILORED MEALS, MEDICALLY TAILORED GROCERIES, AND PRODUCE PRESCRIPTIONS. FIM treatment options for individual patients vary in intensity and level of support. More robust targeted treatments, such as medically tailored meals (MTMs), are typically provided to patients with multiple complex comorbidities and limited activities of daily living; and less intensive interventions, such as medically tailored groceries and produce prescriptions, are provided to larger populations of less ill individuals with a diet-sensitive condition or risk factor (Table 1). All FIM programs typically incorporate registered dietitian nutritionists (RDNs) for tailoring of food to particular medical conditions and patient preferences, social workers to help identify and refer patients with specific social needs, and, increasingly, the use of electronic tools for program screening, referral, and implementation. Nutrition education (focusing on information about foods and nutrients to increase or minimize for health) and culinary education (addressing how to shop for and prepare healthy foods) are common supportive components, delivered in person or digitally. In programs to date, the systematic training of physicians and other clinicians in using FIM has been variable. As discussed subsequently, national calls are growing for appropriate MNE,[30] with some initial response from medical and other health professional schools, health-related societies, licensing boards, and accreditation councils.

PATIENT ELIGIBILITY. Patients are generally identified based on the combination of: 1) having at least 1 diet-sensitive condition, especially related to cardiometabolic health; and 2) having 1 or more social needs (eg, food insecurity, financial strain, homelessness). Commonly targeted health conditions include diabetes, hypertension, heart failure, and high-risk pregnancy; others include chronic kidney disease, cancer, and HIV. Notably, FIM interventions are generally used for treatment to improve disease management and reduce complications rather than for long-term prevention. Sometimes, the presence of a diet-sensitive disease condition is the only formal eligibility criterion (ie, without assessing social needs).[31,32] This places FIM programming more squarely in the category of a medical treatment rather than a social needs intervention. Yet most FIM programs also emphasize health equity by incorporating an approach to reach vulnerable patients, such as targeting individuals insured through Medicaid or receiving care in a federally qualified community health center.

INTERLINKAGES OF HEALTH SYSTEMS WITH GOVERNMENT NUTRITION PROGRAMS. Health sector strategies to increase enrollment in federal nutrition programs can support FIM efforts (Figure 1). These include the Supplemental Nutrition Assistance Program (SNAP; formerly Food Stamps), Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), school meals, and meals for the elderly. Many FIM programs aim to support enrollment into these programs through referrals to local eligibility offices, assisting with application completion, or embedding an eligibility worker to conduct on-site application submissions. For example, in 2019, Kaiser Permanente launched Food for Life, a major text-messaging campaign that helped 120,000 members recognize SNAP eligibility and complete the application process. Coenrollment into multiple programs also holds promise, such as into Medicaid and SNAP, owing to similar eligibility and enrollment requirements. Yet, lack of currently compatible technologies (eg, EHR vs SNAP systems) has made these initiatives less successful, though with some exceptions, eg, certain EHRs allow referrals to WIC. Coenrollment is also challenging when enrollment criteria differ, such as for FIM, where eligibility is generally based on a disease condition and social needs, vs federal nutrition programs, where eligibility is often income-based; this must be addressed to maximize patient benefits.

FIM services and federal nutrition programs should not be considered as being redundant. The former represent medical treatments for disease conditions, generally do not have income criteria, and provide nutritionally tailored meals/foods. Conversely, federal nutrition programs are income based, target various segments of the general population, and except for WIC and school meals, do not have major nutritional criteria. Both FIM and federal nutrition programs are supplemental, ie, supporting partial food benefits rather than a household's entire nutritional needs. Therefore, access to multiple programs is often needed and they can act synergistically. In addition, because of differences in eligibility criteria, many people will qualify for only a subset of available programs.

In many ways, WIC can be considered as an original FIM program, because it requires health care provider referral, adult eligibility is based on a specific condition (pregnancy), and benefits provide access to specific healthy food items and nutrition education. The fact that WIC has been successfully scaled into every U.S. county and has shown positive impacts on

JACC VOL. 83, NO. 8, 2024
FEBRUARY 27, 2024:843-864

TABLE 1 Examples of Food Is Medicine Strategies to Address Food and Nutrition Security, Health, and Health Equity

	Target Population	Intervention
Medically tailored meals[a]	• People with severe complex chronic conditions that limit activities of daily living and cause high burdens of disability, illness, and health care utilization, such as heart failure, poorly controlled diabetes, end-stage renal disease, acquired immunodeficiency syndrome, or cancer • People with other high-risk conditions, such as high-risk pregnancy	• Fully prepared meals, tailored to the medical needs of the recipient by an RDN • Typically provided as 10 weekly meals (eg, lunch and dinner on weekdays), free and delivered to the home • Generally combined with individual, group, or digital nutrition and culinary education • Variable duration, but often 3-6 months with interim reassessments as well as possibility of renewal
Medically tailored groceries[a]	• People with 1 or more major complex diet-related health risks or conditions, such as diabetes, but who can still prepare and cook their own meals	• Healthy food items that are preselected, often by an RDN or other qualified professional, and provided to eligible patients • Items typically include a selection of free fruits, vegetables, beans, and whole grains, with or without nuts/seeds, poultry, fish, or dairy. Items may be fresh, frozen, or canned. • Implementation is most often by home delivery or direct pick-up at a food bank or health care facility (eg, a brick-and-mortar "food farmacy" in a federally qualified health care center). • Generally combined with individual, group, or digital nutrition and culinary education • Variable duration, but often 3-6 months with interim reassessments as well as possibility of renewal
Produce prescriptions[a]	• People with at least 1 diet-sensitive health risk or chronic condition, such as diabetes, prediabetes, hypertension, obesity, or coronary artery disease, who can still shop, prepare, and cook their own meals • Pregnancy and early life (first 1,000 days)	• Discounted or free produce, typically fresh fruits and vegetables • Benefits provided by electronic debit cards or paper vouchers, which are redeemable at grocery stores or farmers markets, picked up in the health care setting, or used for home delivery • Generally combined with individual, group, or digital nutrition and culinary education. • Variable duration, but often 3-6 months with interim reassessments as well as possibility of renewal
Government nutrition security programs	• Adults and children from lower-income households (eg, a family income-to-poverty ratio of <1.3 for SNAP and <1.85 for WIC) • Individuals also often have food or nutrition security as well as other social needs, although these are not formally assessed for screening or eligibility	• Screening, connecting, and supporting enrollment of appropriate individuals into federal nutrition programs • Programs include SNAP for the general population, WIC for pregnant mothers and children through age 5 years, school breakfast and lunch programs for children in grades K-12, and nutrition programs (eg, meals on wheels) for older adults. • Programs have variable nutritional standards: specific healthy food and nutrient standards for WIC, school meals, and meals on wheels, and no similar standards for SNAP • Whereas community and public health organizations have long aimed to increase enrollment among eligible individuals, health care systems are now increasingly involved in direct screening, referral, and enrollment support for patients in their care
Population-level healthy food programs and policies	• Adults and children within the general population, with or at risk for poor metabolic health	• Programs and policies to address systems and environmental barriers to equitable healthy food in populations—examples: • Child care: stronger nutrition standards for school meals and early child care systems, such as through federal policy, state licensing, or quality ratings • Procurement: leveraging city and state bulk food procurement policies and practices to prioritize more nutritious foods; and strengthening nutrition standards within food service, including in municipal buildings, park/recreation centers, workplaces, hospitals, universities, and charitable food networks • Information: package labeling, restaurant menu labels, package warning labels • Employer wellness: benefits programs including nutrition education and incentives for healthier eating • Fiscal: excise taxes or other disincentives for unhealthful foods or beverages; incentives for healthier foods • Regulatory: standards or restrictions on sodium, added sugar, and other additives

[a]Individuals with specific medical conditions are identified by clinicians or other health system staff including RDNs, social workers, and community health workers, who screen and refer eligible patients to appropriate services as part of their treatment plan. These programs often, but not always, also focus on more vulnerable populations at risk for health disparities, such as those with low incomes, food insecurity, and other identified social needs. Adapted from Downer et al,[21] Mozaffarian et al,[23] and Mozaffarian.[28]

RDN = registered dietitian nutritionist; SNAP = Supplemental Nutrition Assistance Program; WIC = Special Supplemental Nutrition Program for Women, Infants, and Children.

numerous short- and long-term health outcomes in both women and children[33] supports the promise of newer FIM programs to be widely scaled to meaningfully improve health.

POPULATION-LEVEL HEALTHY FOOD POLICIES. At the base of the FIM pyramid are population-level healthy food programs (Figure 1). These recognize that access to convenient, affordable, and nutritious

foods cannot be fully addressed by health care or individual choice, but is influenced by home and work environments. Several evidence-based policy, systems, and environmental interventions can advance nutrition security and health equity in communities (Table 1). For example, the Centers for Disease Control and Prevention (CDC) supports healthier community food options through guideline development, surveillance, and monitoring, support for programming, and technical assistance for local organizations and national partners. Other population-level policy strategies include tax credits to support healthier food access within small stores and joint food tax-subsidy programs, such as used in King County, Washington, where revenue from sugar-sweetened beverage taxation supports access to nutritious foods in low-income neighborhoods.

EVIDENCE FOR EFFECTIVENESS

IMPACT OF FIM PROGRAMS ON FOOD SECURITY AND DIET QUALITY. Although FIM interventions have varied in design, target population, intervention, and implementation characteristics, evidence suggests these programs have positive impact on food security and dietary quality. In one systematic review, 3 of 5 produce prescription interventions identified significant pre/post improvements in food security, and a fourth found improvements in most participants but did not report statistical significance.[34] In a separate analysis, individual-level data were pooled from 3,881 participants across 22 newly evaluated produce prescription sites in 12 U.S. states.[32] The programs provided a median of $63/month to purchase fruits and vegetables at grocery stores or farmers markets. At baseline, 56% of households reported food insecurity. After an average intervention duration of 6 months (range 4-10 months), the odds of being food insecure decreased by one-third (OR: 0.63; 95% CI: 24%-48%). These impacts are important because food insecurity can create tradeoff decisions between food, medications, and health care visits.

Although direct provision of MTMs would likely improve food security, few MTM studies have evaluated this endpoint. In a small cross-over randomized trial among patients with diabetes at risk for food insecurity, MTMs for 12 weeks reduced food insecurity from 62% to 42% ($P = 0.047$).[35] Similarly, a small, pre/post intervention of MTMs for 6 months among patients with HIV and/or diabetes showed substantial reduction in food insecurity, from 90% to 46% ($P < 0.0001$).[36]

FIM programs improve diet quality. In a systematic review, 21 of 22 produce prescription studies identified improvements in fruit and vegetable intake.[37] In one meta-analysis, produce prescriptions increased fruit and vegetable intake by 0.8 servings/day (95% CI: 0.2-1.4).[31] In a pooled analysis of new produce prescriptions studies across 12 states, fruit and vegetable consumption increased by 0.85 serving/day (95% CI: 0.68-1.02) in adults and 0.26 servings/day (95% CI: 0.06-0.45) in children.[32] These were pre/post interventions, without randomization or separate control groups, limiting strength of causal inference. Few studies evaluated substitution, ie, how healthy foods replace unhealthy foods. One small pre/post study of produce prescriptions to treat hypertension suggested reduced intake of fast food.[38]

MTMs also improve diet quality, including higher intakes of protective foods and nutrients and (through substitution) lower intakes of unhealthy dietary factors. In a small cross-over randomized trial of MTMs in diabetes, diet quality markedly increased, with mean Healthy Eating Index score improving from 40 to 71 ($P < 0.0001$), based on more fruits, vegetables, and whole grains and fewer solid fats, alcohol, and added sugar.[35] Quasiexperimental studies suggest similar improvements in dietary quality with MTMs, including more fruits and vegetables and fewer salty snacks and sugar.[36,39,40]

IMPACT OF FIM PROGRAMS ON HEALTH OUTCOMES. In a meta-analysis of pre/post produce prescription interventions published through January 2020, hemoglobin (Hb) A_{1c} decreased by 0.8 points (95% CI: 0.1-1.6 points; 5 studies), and body mass index (BMI) by 0.6 kg/m^2 (95% CI: 0.2-1.1 kg/m^2; 3 studies).[31] No significant changes were observed for blood pressure (4 studies) or blood lipids (2 studies). However, the numbers of studies and patients identified were small, and analyses did not consider baseline risk, eg, presence or absence of diabetes, hypertension, hyperlipidemia, or adiposity.

In a more recent patient-level analysis of 11 previously unreported produce prescription interventions (pre/post over an average of 6 months), HbA_{1c} declined 0.3% (95% CI: 0.2%-0.4%) among adults with diabetes (0.6% [95% CI: 0.4%-0.8%] if baseline HbA_{1c} >8.0%), BMI by 0.4 kg/m^2 (95% CI: 0.1-0.6 kg/m^2) among adults with overweight or obesity (by 0.5 kg/m^2 [95% CI: 0.2-0.9 kg/m^2] if baseline obesity), and systolic and diastolic blood pressure by 8 mm Hg (95% CI: 7-10 mm Hg) and 5 mm Hg (95% CI: 4-6 mm Hg), respectively, among adults with hypertension (by 11 mm Hg [95% CI:

8-14 mm Hg] and 9 mm Hg [95% CI: 7-12 mm Hg], respectively, if baseline blood pressure ≥140/90 mm Hg).[32] Among 1,817 children aged 2 to 17 years in these programs, child BMI *z*-score did not significantly change, although program durations averaged only 6 months. From baseline, self-reported health status improved, with higher odds of improving 1 level in self-reported health among both adults (OR: 1.62; 95% CI: 1.30-2.02) and children (OR: 2.37; 95% CI: 1.70-3.31).[32] Another small produce prescription study in children identified decreased need for oral antibiotics.[41]

In a small pilot randomized trial among 112 patients with diabetes at a federally qualified health center, an 8-week produce prescription program reduced HbA1c by 0.5 points in the intervention group ($P = 0.006$), with no change in the control group ($P = 0.89$).[42] In another randomized trial in patients with diabetes, an on-site food pantry-based grocery program did not significantly reduce HbA1c at 6 months compared to control,[43] although only one-half of the weekly grocery allotments were picked up, drop-out and missing data were high (30%), and groceries included poultry, dairy, and grains, which would not be expected to improve HbA1c.[7] In the initial report of a third randomized trial, a home-delivered produce prescription program of fruits, vegetables, beans, nuts, and plant oils in Medicaid-insured patients with diabetes reduced HbA1c by 0.37 points compared to control ($P = 0.028$).[44]

Fewer MTM studies have assessed health outcomes or disease management. In exploratory analyses of a short-term (10-week) randomized trial among 1,977 patients with heart failure, diabetes, or chronic kidney disease, MTMs were associated with lower mortality.[45] In another randomized trial, diabetic patients reported less frequent hypoglycemia on MTMs (47% vs 64%; $P = 0.03$).[35] In a small pre/post study, MTMs improved diabetes distress and perceived diabetes self-management ($P < 0.01$ each) and adherence with antiretroviral therapy in patients with HIV (from 47% to 70%; $P < 0.05$).[36] Participation in MTMs has also been associated with fewer days of poor mental health (19% vs 32% of days; $P = 0.03$),[35] decreased depressive symptoms and binge drinking, and less frequent sacrificing of food for health care or prescriptions.[36]

As noted above, many of these studies are quasi-experimental or pre/post interventions, with fewer randomized trials. Thus, benefits on health outcomes appear promising, and larger, longer-term trials are needed. Multiple randomized trials are now ongoing, including in diabetes, cancer, and high-risk pregnancy (eg, NCT04986670, NCT04866823, etc).

IMPACT OF MNE FOR PHYSICIANS AND OTHER CARE PROVIDERS. An effective FIM strategy requires clinicians who can identify, educate, and counsel those with dietary risk, and apply evidence-based strategies to address food and nutrition security. Effectiveness of clinician-delivered dietary counseling is well supported by existing literature, with even brief counseling leading to improved diet quality.[46] Numerous medical organizations recommend dietary counseling by clinicians and other health care team members.[18] A 1985 National Academy of Sciences report recommended at least 25 hours of MNE, a threshold met by few medical schools. Despite ensuing efforts, national surveys have shown decreases in MNE teaching hours over time.[47] MNE to build competencies and skills has also lagged. A recent systematic review confirmed consistent lack of substantive MNE in medical schools across continents.[48] These training gaps are misaligned with the 2010 Affordable Care Act's call for stronger emphasis on social and behavioral determinants of health, and with health sector recommendations in the 2022 National Strategy on Hunger, Nutrition, and Health.[16,17] Accordingly, studies consistently document that physicians in training and practice recognize the importance of nutrition for patient care but lack sufficient MNE, knowledge, and competencies to assess, advise, or treat patients via diet counseling or other nutrition-based interventions.[48-50] These deficiencies contribute to low rates of diet counseling in clinical practice[51] and public confusion about nutrition and health.

Several studies have evaluated effectiveness of MNE interventions. A 2019 systematic review identified 24 studies published between 2012-2018, including 11 U.S. studies, that investigated MNE provided to medical students or recently trained physicians.[48] These studies confirmed baseline gaps in MNE that contributed to physician concerns about ability to engage patients in dietary education and counseling. Five studies (4 from the United States) evaluated effectiveness of MNE curriculum initiatives. These demonstrated improvements in clinician competencies for patient education and dietary assessment, personal nutrition knowledge, and confidence around clinical and public health aspects of nutrition.

A newer approach termed "culinary medicine" connects didactic MNE with hands-on food and cooking experiences. A small randomized trial found culinary medicine improved both medical student competences and blood pressure and blood cholesterol control in their patients.[52] A prospective, multisite cohort study of 4,026 students across 45 U.S. medical schools examined nutritional competencies

and personal dietary habits of students exposed to culinary medicine, compared to propensity-matched students receiving a more traditional curriculum.[53] With follow-up to 5 years, culinary medicine training was associated with higher achievement in 25 diet-related preventive cardiology competencies (combined OR: 2.14; 95% CI: 2.00-2.28) as well as improved personal dietary habits (eg, lower soft drink consumption [OR: 0.56; 95% CI: 0.37-0.85] and adherence to a Mediterranean-type diet [OR: 1.40; 95% CI: 1.07-1.84]).[53]

A recent meta-analysis of varied MNE programs, including some culinary medicine, examined diet competencies across 1,698 trainees in 11 studies.[54] Competencies improved overall (effect size = 9.80; 95% CI: 7.15-12.45), but with variable study quality according to STROBE criteria. Two high-quality studies had a higher effect size than 3 moderate-quality or 6 low-quality studies, with percentage improvements in medical student competencies for counseling patients on nutrition of 41% ± 4%, 18% ± 13%, and 21% ± 14%, respectively.[54]

In summary, the evidence from mostly nonrandomized interventions has limitations but suggests that MNE can be effective and scalable, and further optimized when including culinary medicine. Downstream effects include improved clinician competency, better clinician dietary habits, and, in limited studies, better patient outcomes. Yet implementation of MNE remains limited by perceived time constraints and lower prioritization in education and training programs. More rigorous methodology and longer follow-up are needed to determine optimal MNE approaches and long-term impacts.[48]

IMPACT OF FIM PROGRAMS ON VULNERABLE POPULATIONS AND HEALTH EQUITY. A primary goal of FIM interventions is to address poor nutrition as a direct determinant rather than a social determinant of health. Consequently, while health sector efforts to address social needs are sometimes considered to be adjacent to medical care, FIM interventions represent a health care-integrated therapy, prescribed by clinicians, to treat diet-sensitive conditions. At the same time, FIM interventions reduce food insecurity, which is a major social determinant of health and marker of poverty, and generally are prescribed to populations suffering from health disparities. Thus, FIM programs show promise to not only improve disease management and health status, but also narrow gaps in food security and health equity. However, while benefits for diet quality, food security, and health have been seen, the effects of FIM on health disparities per se, and the comparative effectiveness of different FIM program designs to address disparities, have not been formally evaluated. This represents an important evidence gap for future research.

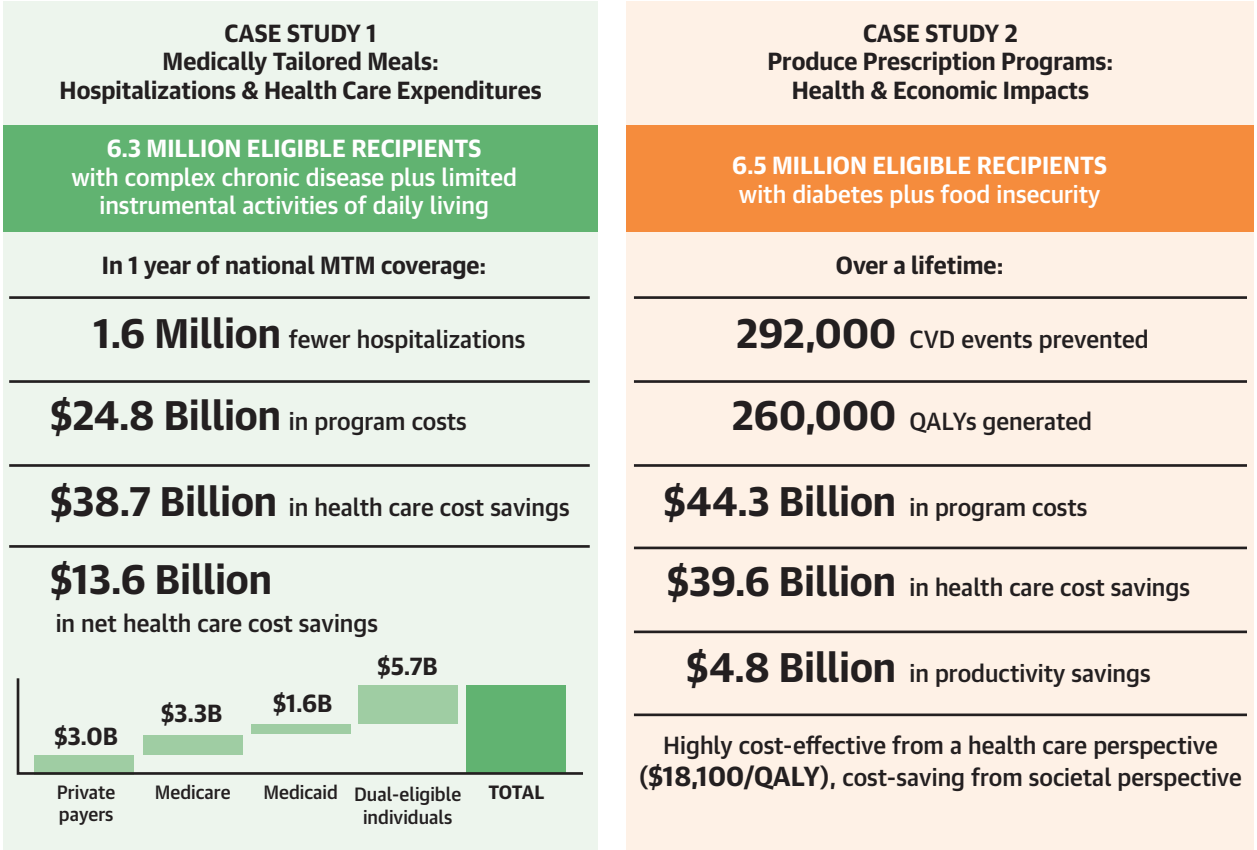
IMPACT ON HEALTH CARE UTILIZATION, COSTS, AND COST-EFFECTIVENESS. Payers and other partners are increasingly mobilized by evidence that addressing poor nutrition and food insecurity through FIM may reduce health care utilization. Quasiexperimental and pre/post interventional studies suggest that provision of MTMs to patients with complex chronic medical conditions reduces emergency department visits, inpatient admissions, lengths of stay, skilled nursing facility admissions, and total medical spending.[55-57] In a meta-analysis of 5 studies, MTM programs had an average implementation and food cost of $9.30 per meal and were associated with 19.7% lower annual health care expenditures (95% CI: 6.9%-32.4%) and 47% fewer annual hospitalizations (95% CI: 31.7%-62.3%).[58] In a national simulation model, 6.3 million U.S. adults could be eligible for MTMs, based on having a major diet-sensitive condition (most commonly CVD, diabetes, or cancer) plus limited instrumental activity of daily living. Based on an intervention of 10 weekly MTMs for 8 months, annual intervention costs were estimated to be $24.8 billion, health care savings to be $38.7 billion (including 1.6 million fewer hospitalizations), and net total cost savings to be $13.6 billion (**Figure 2**).[58,59]

In randomized trials, one small short-term (4-week) MTM intervention suggested trends toward fewer 30-day heart failure readmissions (11% vs 27%; $P = 0.06$) and total days of rehospitalization over 12 weeks (17 vs 55; $P = 0.055$).[60] In exploratory analyses of another short-term (10-week) randomized trial in heart failure, MTMs did not reduce 90-day all-cause hospitalizations, but were associated with fewer heart failure hospitalizations.[45] The short treatment duration in these trials (4-10 weeks) may have limited effects on health care utilization compared with the longer treatment durations (mean 8 months) in the quasiexperimental and pre/post interventional studies.[58]

Compared with MTMs, fewer studies have assessed effects of medically tailored groceries or produce prescriptions on health care utilization. A quasiexperimental analysis of a produce prescription program in Hartford, Connecticut, found no significant impact on health care utilization or clinical outcomes compared with matched control subjects, but the program was implemented at the start of the COVID-19 era when health care was highly disrupted.[61] In a

JACC VOL. 83, NO. 8, 2024
FEBRUARY 27, 2024:843-864

FIGURE 2 Cost-Effectiveness of Implementing Food Is Medicine Programs in the United States



The health and economic impacts were estimated for (left) a national program of medically tailored meals among adults with 1 or more complex nutrition-sensitive chronic conditions and 1 or more limitations in activities of daily living, and (right) a national program of produce prescriptions for adults with type 2 diabetes and food insecurity. Each model was populated with nationally representative data and the best available evidence on effects and costs of these programs. Reproduced with permission from Deuman et al.[59] CVD = cardiovascular disease; QALY = quality-adjusted life year; MTM = medically tailored meal.

randomized trial among patients with diabetes, a 6-month on-site food pantry-based FIM program did not significantly reduce total health care utilization, although lower costs of inpatient and emergency department claims appeared offset by higher costs of more outpatient visits associated with weekly on-site grocery pick-up.[43] Compared with MTMs, produce prescription programs are often "low dose" (eg, $30-$50/month in benefits over 3-6 months) and target lower-risk populations. Given this lower intensity and baseline risk, produce prescriptions may require more extended periods (eg, ≥1 year) to detect meaningful changes in health care utilization. In a national modeling study, the potential health and economic impacts of produce prescriptions were assessed for adults with both diabetes and food insecurity (~6.5 million Americans).[59,62] Over a lifetime, the program was estimated to prevent 292,000 cardiovascular events, cost $44.3 billion to implement, and save $39.6 billion in formal health care expenditures and $4.8 billion in productivity costs (**Figure 2**). From a health care perspective, the program was highly cost-effective ($18,100 per quality-adjusted life-year gained), and from a societal perspective it was cost-saving.

Thus, current evidence suggests that MTMs in high-risk, high-utilization patients may be cost-saving, and produce prescriptions in more general patients with diet-related conditions may be highly cost-effective (or cost-saving from a societal perspective). More quasiexperimental, randomized, and simulation research is needed to determine how

FIM programs can support better health and reduced health inequities as high-value care.

NEW NATIONAL AND LOCAL PROGRAMS AND POLICIES

As evidence on the benefits of FIM interventions has grown, so has interest among payers and policy makers. New opportunities have emerged and are accelerating to incorporate FIM programs into health care delivery and financing (Table 2).

FIM IN MEDICAID. As the main U.S. safety-net health insurance program, Medicaid provides coverage to around 90 million low-income individuals. FIM interventions have (with limited exceptions) historically not been included in Medicaid benefits,[63] but in recent years, federal and state Medicaid officials have explored policy flexibilities to address that gap. One of the most prominent is Medicaid Section 1115 Demonstration Waivers, which are 5-year waivers that allow states to experiment with new approaches to eligibility, benefits, and payments.[63] At the end of 2022, the Centers for Medicare and Medicaid Services (CMS) released a framework clarifying use of Section 1115 waivers to provide nutrition supports including FIM. As of July 2023, 7 states (Arkansas, California, Massachusetts, New Jersey, North Carolina, Oregon, and Washington) have received CMS approval to implement Section 1115 waivers including FIM interventions, and another 4 (Delaware, Illinois, New Mexico, and New York) await approval.[64] Medicaid officials have also expanded flexibilities within Medicaid Managed Care (ie, provision of Medicaid services via private health plans) to increase access to FIM. In 2023, CMS released new guidance[65] and proposed regulations[66] describing how states can utilize "in lieu of services" to cover FIM services as cost-effective substitutes for more traditional care. California is combining 1115/1915(b) waivers and in lieu of services authority to allow its managed care plans to cover 14 categories of "community supports," including FIM interventions.[67]

FIM IN MEDICARE. Like Medicaid, Medicare—the U.S. public health insurance program for individuals aged ≥65 years or with disabilities—has historically not provided coverage for FIM services. Recent policy changes have created opportunities within Medicare Advantage (ie, Medicare Part C, which is administered by private health plans). Previous provisions limited home-delivered meals to certain enrollees for a short duration, but in 2018, Congress expanded this provision with Special Supplemental Benefits for the Chronically Ill (SSBCI). This allows Medicare Advantage plans to cover a spectrum of nutrition services, including MTMs for longer durations as well as food and produce. In a recent analysis, 422 plans now provide coverage of meals as part of SSBCI and 929 plans provide coverage of food and produce.[68] This represents a marked increase from the program's launch in 2020, when only 71 and 101 plans, respectively, provided SSBCI coverage for these services.[68] CMS has created other opportunities to support FIM access through large-scale Medicare demonstrations, including Value-Based Insurance Design, which requires participating plans to provide supplemental benefits responding to health-related social needs,[69] and a new Advance Investment Payments option within the Medicare Shared Savings Program to provide new accountable care organizations with up-front funds that can cover FIM services.[70]

These Medicare and Medicaid innovations to expand access to FIM rely on managed care, accountable care organization, or demonstration project authorizations. The standard Medicare Parts A

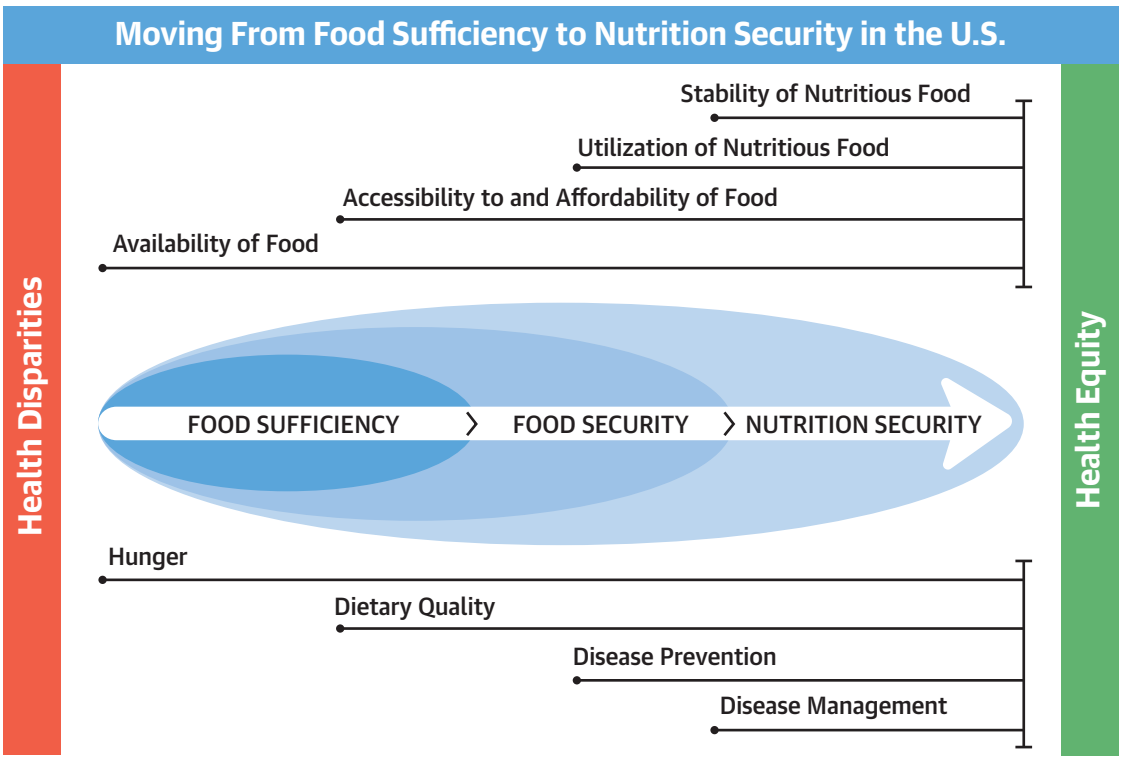
TABLE 2 Examples of Accelerating Activities, Programs, and Opportunities Around Food Is Medicine

Coverage for FIM interventions in state Medicaid 1115 and 1915(b) waivers, Medicaid managed care organization "in lieu of," value-added, or quality improvement services
New FIM interventions in Medicare Advantage Organization and Medicare Shared Savings Accountable Care Organization programs
Private health care initiatives in FIM, eg, Kaiser Permanente, Geisinger Health, Elevance, Centene, others
The 2022 White House National Strategy on Hunger, Nutrition, and Health, including a core pillar on integrating nutrition and health that prioritizes many FIM commitments and actions
Veterans Affairs and Indian Health Services pilot programs on produce prescriptions
Center for Medicare and Medicaid Innovation consideration of a demonstration project on medically tailored meals
Expansion and evaluation of the U.S. Department of Agriculture Gus Schumacher Nutrition Incentive Program produce prescription programs
Integration of FIM into Centers for Disease Control and Prevention community-based programs such as SPAN, HOP, and REACH
Progress toward mandatory universal screening for food insecurity in all federal health care systems
American Academy of Pediatrics and American College of Lifestyle Medicine commitments on FIM training for physicians
ACGME plans to institute mandatory nutrition education in graduate medical education by 2026
Joint ACGME, AAMC, and AACOM summit on nutrition education in undergraduate medical education
National Institutes of Health Office of Nutrition Research proposal for new FIM Centers of Excellence
Rockefeller Foundation/American Heart Association joint commitment to raise $250 million for FIM research
Nonprofit coordination and collaboration on FIM, including the Food Is Medicine Coalition and National Produce Prescription Collaborative
Private-sector FIM innovations from companies such as Instacart, Kroger, Season Health, Mom's Meals, and others

AACOM = American Association of Colleges of Osteopathic Medicine; AAMC = Association of American Medical Colleges; ACGME = Accreditation Council for Graduate Medical Education; FIM = Food Is Medicine; HOP = High Obesity Program; REACH = Racial and Ethnic Approaches to Community Health; SPAN = State Physical Activity and Nutrition Program.

JACC VOL. 83, NO. 8, 2024
FEBRUARY 27, 2024:843-864

FIGURE 3 Intersections of Food and Nutrition Security With Health Equity



The transition from food sufficiency to nutrition security (arrows) requires a spectrum of policies and programs from those that facilitate the fulfillment of basic human needs such as hunger and food availability to those that achieve utilization and stability of nutritious food for disease prevention and management. Food security requires food sufficiency (overlapping darker circles), and nutrition security requires both food sufficiency and food security (overlapping lighter circle). Given the central role of nutrition-related conditions in health disparities, this transition is critical for individuals and populations to shift away from health disparities and toward health equity. Adapted with permission from Thorndike et al.[26]

and B program—covering two-thirds of enrollees—currently has no flexibilities to cover FIM. In June 2023, a bipartisan bill was introduced in the Senate (S.2133) directing the Secretary of Health and Human Services to conduct a 4-year demonstration program to provide MTMs in Medicare Parts A and B and evaluate effects on health and health care utilization. In addition, CMS is advancing policies to provide incentives for or require universal screening for food insecurity in Medicare,[71] which could drive uptake of FIM programs, policies, and payment opportunities.

FIM IN PRIVATE PAYER AND INTEGRATED HEALTH SYSTEMS. Integrated health systems that practice accountable care and population health management have been early adopters of FIM interventions in at-risk members. Other private payers are also now initiating FIM interventions in high-risk beneficiaries. All generally target cardiometabolic risks such as diabetes or obesity, as well as other priority groups such as high-risk pregnancy. Most of these programs were initiated as small pilots that were then broadened and systematized, providing early evidence of scalability across diverse health delivery systems. A few relevant examples are summarized here.

In 2011, Kaiser Permanente's outpatient clinics in Colorado began screening patients for food insecurity and connecting them to community-based organizations to assess and support eligibility and enrollment in federal and local food assistance.[72] Early lessons were that food insecurity was much higher than initially suspected, patients were more likely to connect to community-based organizations when the process was active and automated, and receipt of services was higher when clinical teams included social workers. This initial program has been scaled and expanded across Kaiser's national footprint, including outreach and social screening of vulnerable populations, such as Medicaid enrollees, an organizational standard of e-screening for food insecurity,

TABLE 3 Screening Tools for Food Security, Nutrition Security, and Diet Quality

Measure	Conceptual Framework	Measurement Tools	Classification	Validity
Food security	• The availability of and access to food that is safe, affordable, and consistent with food preferences	• The most common screening tools are the 2-item Hunger Vital Sign and the 6-item Food Security Survey Module, each validated against the criterion-standard 18-item USDA assessment.[11] • Food security is generally assessed at the household level.	• High, marginal, low, or very low food security • Low and very low food security are combined to report overall food insecurity. • Food insecurity is not synonymous with hunger, a downstream potential consequence of food insecurity that manifests as disrupted food intake and reduced calories.[a]	• Two decades of research and evaluation have shown that measures of food security are associated with disease conditions, mental health, academic achievement, and health care utilization.[10] • Individuals experiencing food insecurity have higher risk of suboptimal nutrition, diet-related diseases, and resulting financial disparities.[82]
Nutrition security	• A newer concept that has been defined as "consistent access, availability, and affordability of foods and beverages that promote well-being and prevent and if needed, treat disease"[25] • Whereas tools to measure and address food security generally focus on sufficiency of calories (quantity), nutrition security focuses on nourishing foods (quality), a complementary concept for advancing health and health equity.[28]	• A 2-item Nutrition Security Screener (NSS) assesses presence and severity of nutrition insecurity as well as potential drivers such as food cost, store locations, transportation, knowledge, time, kitchen equipment, physical limitations, taste, and traditional preferences.[28] • Another 4-item nutrition security survey asks about consumption and respondent concerns around foods for health and well-being.[85] • Nutrition security is generally assessed at the household level.	• Optimal classification schemes require more research and validation in diverse populations.[28]	• Early findings suggest that nutrition security provides new, complementary information compared with assessment of food security alone, is associated with self-reported physical and mental health, and is higher in certain income, race, ethnicity groups (Dariush Mozaffarian, personal communication, July 21, 2023). • According to the USDA, "Nutrition security builds on food security, emphasizing the coexistence of food insecurity and diet-related diseases and disparities."[81]
Diet quality	• A complementary concept that encompasses the nutritional quality of the foods, beverages, nutrients, or overall dietary patterns consumed • Diet quality can be considered as a key downstream mediator of effects of food security, nutrition security, and FIM interventions.	• Detailed 24-hour recalls and food frequency questionnaires are valid but relatively time intensive. • For clinical screening and FIM programs, short diet quality screeners are more practical, generally focusing on key food groups (eg, fruits, vegetables, fast foods, sugary beverages, etc.) in varying length and detail.[86] Examples include the 21-item Dietary Screening Questionnaire[87] and the shorter Mediterranean Diet as well as others.	• Although diet quality screeners do not delineate a complete measure of dietary habits, the selected relevant food groups make them a reasonable proxy for overall diet quality. • Measures of diet quality should generally be assessed, along with food and nutrition security, in populations being considered for or receiving FIM services.	• Both detailed diet quality assessments and screeners have been validated against major disease outcomes. • Several validated rapid diet screener tools are available, meeting theory and practice requirements, taking <5 minutes to complete, and integrated into EHRs for actionable feedback.[86]

[a]Formerly, USDA termed low food security as "food insecurity without hunger" (defined as reduced quality, variety, and desirability of household diets, but with food quantity and eating patterns not substantially disrupted), and very low food security as "food insecurity with hunger" (defined as disrupted eating patterns and reduced food intake).

EHR = electronic health record; FIM = Food Is Medicine; USDA = United States Department of Agriculture.

active and automated referral processes, and a corresponding national policy engagement strategy. In 2022, Kaiser Permanente committed $35 million to FIM programs, including a pilot of medically tailored groceries showing improvements in healthy eating, weight management, and HbA_{1c},[73] an ongoing randomized controlled trial of medically tailored groceries in 450 patients with diabetes and elevated HbA_{1c},[74] and additional planned research, scaling, and evaluation.

Geisinger Health is targeting patients with diabetes and elevated HbA_{1c} through a FIM program using brick-and-mortar "food farmacies" in 3 clinical centers and supported by the charitable food system, grants, the health system, and private donors. Patients receive a prescription to pick up groceries plus recipes for at least 10 meals per week for themselves and their household. The program includes 20 hours of nutritional and diabetes care education and self-management support through a mobile-app. In a pre/post analysis, mean HbA_{1c} levels decreased from 9.6% to 7.5% after 18 months on the program, with a substantial decrease in net annual health costs.[75] However, a step-wedge randomized trial of this food-pantry based program did not show reductions in health care utilization.[43]

Blue Cross/Blue Shield is piloting multiple FIM programs in Minnesota, North Carolina, California, and other states. In North Carolina, for example, a 6-month pilot program evaluated grocery box deliveries ($60 twice per month) combined with recipes and weekly goal-directed health coaching for low-

JACC VOL. 83, NO. 8, 2024
FEBRUARY 27, 2024:843-864

TABLE 4 Challenges for Undergraduate and Graduate Medical Nutrition Education

Lack of curricular guidance resulting in variations in the quality and quantity of nutrition education
Limited curricular hours in the medical education curriculum
Lack of standardized methods and assessments to evaluate nutrition knowledge and skills
Limited number of physician nutrition specialists that serve as role models
Limited nutrition services or clinics that offer a collaborative and interprofessional training environment for medical students, residents, and fellows
Conflicts between previous nutrition knowledge and emerging new scientific insights and clinical evidence
Lack of institutional commitment to making nutrition education a priority
Lack of payment structures to support nutrition counseling by physicians or registered dietitian nutritionists
Lack of universal competencies, such as across the 6 established ACGME learning domains

ACGME = Accreditation Council for Graduate Medical Education.

income members with diabetes.[76] Pre/post analyses at 6 months demonstrated >80% program satisfaction, an 18% absolute decrease in food insecurity, 2-point drop in BMI, 9% decrease in obesity, and improved physical activity and mental health scores. A cost analysis found reduced total monthly medical costs of $139 per member (accounting for the cost of the FIM program) and increased medication adherence. Extrapolating the program to all eligible members, the investigators estimated annual net cost savings of $8.5 million to $13.1 million.

Elevance Health (including Anthem and other health companies) has an active FIM strategy, investing more than $30 million in food and nutrition security programming and evaluation through its foundation.[77] For example, a recent $14 million grant supports 30 member food banks working with health care partners to determine effects of FIM services on clinical outcomes in patients with diabetes, high cholesterol, and high blood pressure. Elevance is also offering MTMs to >100,000 plan participants with specific disease conditions in Florida through its Chronic Meal Benefit; they can receive 42 meals on hospital discharge and 180 meals annually through wellness centers and home delivery. Aetna has partnered with community organizations in Florida on a Mini Mobile Food Farmacy and has integrated MTMs into its Medicare Advantage Plan for individuals with highest dietary risk. Promedica, a nonprofit health care system in Ohio, has built and supports a full-service grocery store in a low-income area of Toledo, offering locally sourced produce, local hiring, and a community hub with job training, nutrition education, and cooking classes.

Together, these payer initiatives provide evidence of broad interest in piloting and scaling FIM services. The experiences to date support continued expansion and testing of FIM programs across governmental and nongovernmental payers for vulnerable populations, especially those with cardiovascular risk factors such as diabetes.

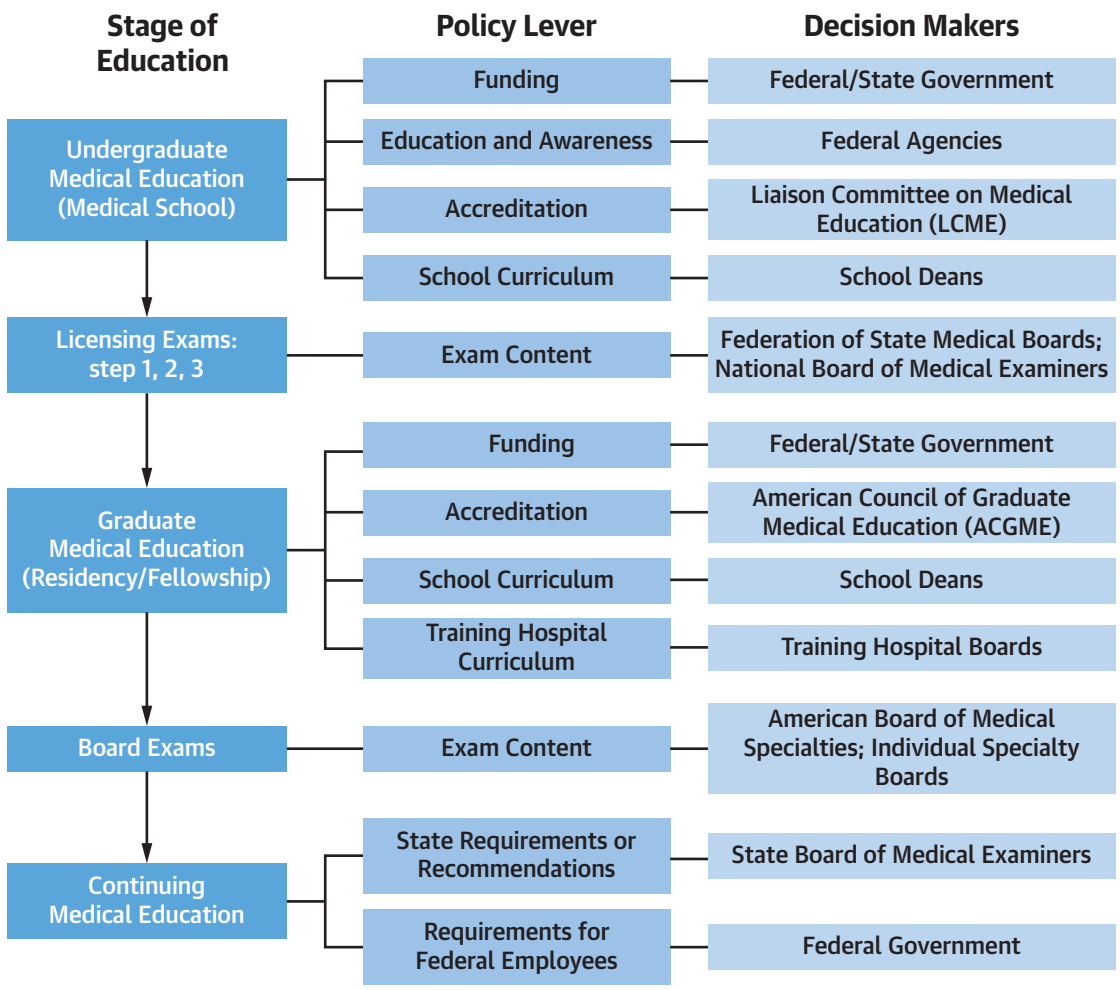
PROVIDERS OF FIM SERVICES

A range of nonprofit, charitable, and faith-based organizations have been in the vanguard of providing FIM.[23] Coalitions such as the Food is Medicine Coalition and the National Produce Prescription Collaborative share best practices, disseminate findings, develop standardized practices and metrics, advocate for supportive policy, and provide technical assistance and training to new players. Leaders in the charitable food network, such as Feeding America and Share our Strength, have released toolkits to facilitate FIM partnerships between regional food banks and health care systems. Nonprofits can experience significant challenges in providing FIM services to health care, such as costs of compliance with privacy requirements. However, such challenges can be overcome when funding sources are more consistent, such as in states supporting FIM services through Medicaid 1115 waivers. Incorporating these community-based organizations into FIM strategies can be beneficial owing to their connections to the community, understanding of local barriers and opportunities, and experience in hosting health screenings, government nutrition program applications, and nutrition and culinary education.

The for-profit sector is increasingly offering FIM services. These include start-up ventures such as Mom's Meals, Foodsmart, Season Health, FarmboxRx, Performance Kitchen, and VitaBowl, among others, often supported through venture capital, and larger food retailers such as Instacart, Kroger, and Walmart. These for-profit vendors can bring scale and efficiency to FIM, but may lack some of the community knowledge, interconnections, and commitments of nonprofit vendors. These trade-offs need to be assessed as FIM grows in scale and scope. To date, most peer-reviewed research on FIM has been generated with nonprofit vendors, and few for-profit companies have published peer-reviewed findings on efficacy, although trials are ongoing.[74]

FIM IN THE 2022 NATIONAL STRATEGY ON HUNGER, NUTRITION, AND HEALTH. In September 2022, the White House released an ambitious National Strategy on Hunger, Nutrition and Health, with goals to end hunger, improve healthy eating and physical activity, and reduce diet-related diseases for all

FIGURE 4 Opportunities to Implement and Accelerate Medical Nutrition Education for Physicians



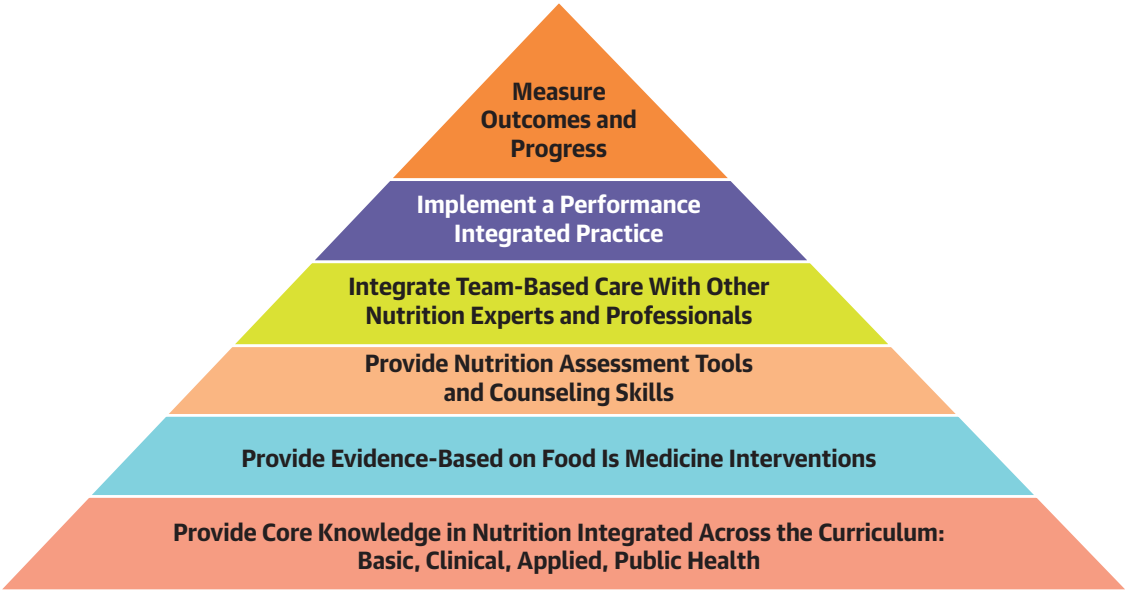
Current medical nutrition education for U.S. physicians is limited. This framework highlights systems changes, including the relevant policy levers and corresponding decision makers, to facilitate the development and maintenance of medical nutrition competencies across the career lifespan (arrows). Adapted with permission from Food Law and Policy Clinic, Harvard Law School.[92]

Americans.[16,17] The Strategy outlined a coordinated national approach to program, policy, and system changes at federal, state, private sector, and civil society levels to advance food and nutrition security and optimal health for the U.S. population. A core pillar of the Strategy is to integrate nutrition and health, prioritizing the importance of nutrition, food security, and FIM interventions in health care for disease management and prevention.

Many specific federal actions were outlined and are being implemented. For example, the National Institutes of Health (NIH) has requested input from all federal agencies on FIM research opportunities, education and training needs, community outreach and engagement, and insurance coverage for FIM services.[78,79] The Department of Veterans Affairs and the Indian Health Service are launching new produce prescription pilot programs. The CDC is supporting FIM interventions through multiple community funding opportunities.[80] The U.S. Department of Agriculture is focusing on advancing food security, nutrition security, and health equity across its 15 nutrition assistance programs.[81] The Health and Human Services Maternal and Child Health Nutrition Program plans to train 30,000 nutrition professionals over 5 years on pediatric obesity prevention, household food security, and nutrition during pregnancy. The NIH plans to invest $140 million to create and support new FIM Networks or Centers of Excellence, building on the successful model of NIH Cancer

JACC VOL. 83, NO. 8, 2024
FEBRUARY 27, 2024:843-864

FIGURE 5 Framework for Medical Nutrition Education to Support Food Is Medicine



The foundation of effective medical nutrition education begins with core knowledge and builds to the development of nutritional assessment tools and integrated team-based care. Ultimately, provider, patient, and health system outcomes should be measured to inform impact and opportunities for improvement.

Centers of Excellence.[78,79] As described above, CMS and various states are pursuing innovations to advance FIM in Medicaid and Medicare. A new interagency working group meets regularly, under the auspices of the White House Domestic Policy Council, to coordinate and advance these federal efforts.

FIM SCREENING FOR FOOD SECURITY, NUTRITION SECURITY, AND DIET QUALITY. As FIM services and MNE advance, practical and efficient clinical screening tools will be required to assess the complementary constructs of food security, nutrition security, and diet quality (**Figure 3**).[23,28] Screening tools should inform the needs of patients and populations, correlates of risk at the patient and environmental levels, and prioritization of interventions, while enabling screening, tracking, benchmarking, and evaluation. For broadest use and effectiveness, screening tools should be incorporated into EHRs, provider education, care pathways, and referral services, and be valid and practical, considering time, workflow, and patient-clinician communication. The need for such screening tools is supported by Department of Agriculture and CDC emphases on food security, nutrition security, and healthful food-based diet patterns.[81-83] A recent Scientific Statement on rapid diet screening tools provides a compelling rationale for widespread adoption.[84]

The EPIC EHR system now includes food security in a social determinants of health module, which is responsive to the 2022 White House National Strategy to incorporate universal food security screening into health care.[17] Ideally, nutrition security and diet quality also should be assessed, given their complementary conceptual and informational facets. Several screening tools are now available to assess these constructs (**Table 3**).[11,25,28,84-87] The broad adoption of EHRs, patient gateways, and mobile apps presents a timely opportunity to incorporate and leverage these screening tools to facilitate evidence-based, equitable FIM services tailored to a patient's needs. This can be supported by current efforts across many EHRs to incorporate screening and referrals for social determinants of health. Ideally, screeners can be completed by patients at home, supplemented by rooming staff at clinic visits, and reviewed by clinicians and other health care team members. Partnerships among clinicians, health care team members, nutrition providers, researchers, and patient representatives will facilitate implementation and assessment.

FIM IN UNDERGRADUATE AND GRADUATE MEDICAL NUTRITION EDUCATION. Successful implementation

of FIM will require a clinician workforce with appropriate MNE. Challenges have inhibited previous meaningful reform (Table 4), but a new multiparty effort is now advancing MNE in the United States. As outlined in a 2012 report, the NIH Nutrition Academic Award helped advance MNE, including educational content, teaching tools, assessment strategies, and exam questions.[88] A follow-up workshop[89] showcased continuing efforts and programmatic recommendations in undergraduate medical education based on new integrated curricula, experiential learning, group interactions, and community outreach activities, and in graduate medical education based on competency-based application of knowledge to patient care, systems-based practice, and practice improvement. New statements by professional organizations call for championing MNE and competency criteria.[90,91] Closing gaps in MNE through FIM initiatives also aligns with recent calls by the Carnegie Foundation, Josiah Macy Jr Foundation, Association of American Medical Colleges, and others to better meet the needs of 21st-century societies and health systems. Recommendations include better training of physicians in population health management and team-based collaborative care, incorporation of behavioral and social sciences into medical school and training curricula, and ensuring competency-based assessment in behavioral counseling.[90,91]

Several U.S. medical schools now meet the recommended minimum of 25 hours of training, not through a single course, but by integrating MNE across the curriculum, reinforcing the influences of nutrition on total body health across the lifespan.[47] A 2019 monograph identified levers to ensure appropriate MNE across training and practice, including funding, accreditation, licensing, curricular changes, and exam content (Figure 4).[92] In 2018, a new bipartisan Congressional Food Is Medicine Working Group[93] met with leadership of medical agency accreditation groups to call for a focus on MNE. In 2021, a resolution was introduced in the U.S. House calling for expanded substantive MNE across health professional training programs.[30] In 2022, the Association of American Medical Colleges, Accreditation Council for Graduate Medical Education (ACGME), and American Association of Colleges of Osteopathic Medicine made a public commitment to the White House to cohost a summit and produce a report on MNE. Held in March 2023, the summit focused on development of MNE clinical experiences and competency-based assessments, including in cultural aspects of counseling, team-based care, and health equity. In April 2023, at a national FIM summit in Boston, the ACGME announced their initiation of the formal process to make MNE a mandatory requirement for residency and fellowship accreditation in the next 2026 update. Learning from other successes, a framework for MNE can support effective provider education around FIM (Figure 5).

TABLE 5 Research Gaps and Opportunities in Food Is Medicine

Optimal study design for different questions, eg, pre/post, natural experiment, quasiexperimental, randomized
Patient eligibility, eg, based on disease conditions, insurance status, clinic location, social needs
Choice between MTMs, medically tailored groceries, and produce prescriptions (including sequential or staged approaches)
Intervention dose, eg, meals per week for MTMs or dollar value for medically tailored groceries and produce prescriptions
Foods covered, eg, fruits and vegetables alone vs other produce; fresh, frozen, or canned; inclusion of eggs, dairy, or meat; flexibility vs preselected items
Cultural inclusiveness of meal and food options
Delivery mechanism, eg, health care pick-up, in-person retail shopping, online retail shopping, home delivery
Duration, especially past 6 months
Food amounts scaled to household size (to account for family sharing) vs for the patient only
Types and extent of nutrition and culinary education, especially comparing RDN-led vs digital, scalable approaches
Effects on health equity
Effects on health care utilization, including cost-equivalence vs cost effectiveness/high-value care vs cost savings
Sustainability following program cessation
Factors that influence process metrics such as screening, referral, enrollment, retention, utilization, satisfaction
Intersections with medical nutrition education for clinicians
Intersections with the EHR, clinical care pathways, and referral systems
Intersections or synergies with other medical treatments, eg, GLP-1 agonists, diabetes prevention program
Differences between FIM vendors, including nonprofit vs for-profit organizations and implications for HIPAA and other structural barriers
In capitated payment systems, comparative effects vs other treatment options on health, health equity, and costs
Optimal payment mechanisms for reimbursing FIM services

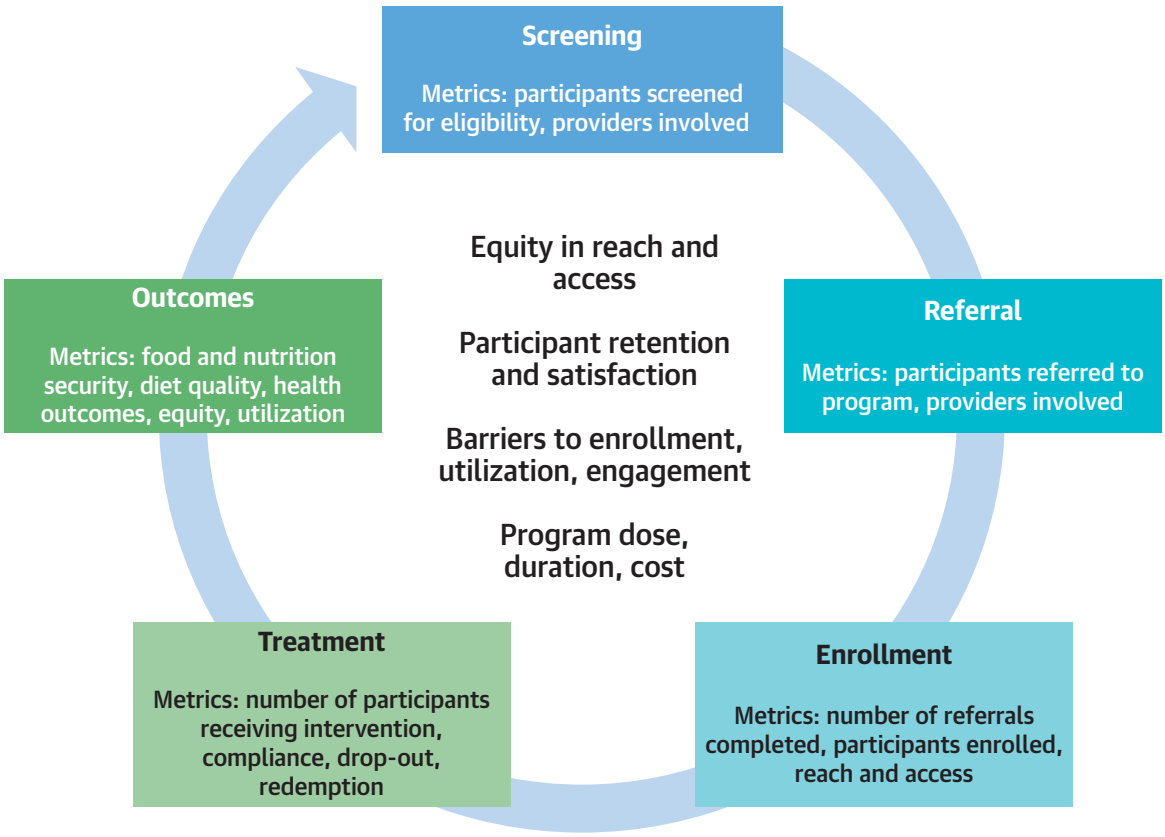
GLP-1 = glucagon-like peptide-1; HIPAA = Health Insurance Portability and Accountability Act; MTM = medical tailored meal; other abbreviations as in Tables 1 to 3.

RESEARCH GAPS AND OPPORTUNITIES

Although FIM interventions show promise and are increasingly deployed across the United States, their optimal scaling and impact require further research (Table 5). Many longer-term studies have used pre/post and quasiexperimental designs (ie, with an external control group), and most randomized trials have had relatively short durations. Larger and longer-term randomized trials are needed. At the same time, an important role may remain in FIM research for quasiexperimental designs, such as stepped-wedge trials, that allow some degree of randomization and a comparator group but also allow all participants to eventually receive an intervention that is expected to have benefit.

JACC VOL. 83, NO. 8, 2024
FEBRUARY 27, 2024:843-864

FIGURE 6 A Framework for Evaluation of Food Is Medicine Interventions



The development of a rigorous iterative scientific framework (arrow) to measure the outcomes of these programs is critical to inform successes and identify opportunities for improvement. Adapted with permission from Levi et al.[23]

Other parameters of interest include patient eligibility, intervention dose, foods covered, cultural inclusiveness, and delivery mechanisms.[94] Duration is especially important to consider. Published studies have ranged from brief (eg, 2-6 weeks) to extended (≥8 months), and CMS Medicaid waivers specify a maximum of 6 months' treatment. Other interventions for chronic conditions, such as many drug treatments for blood cholesterol, blood pressure, obesity, diabetes, or neurodegenerative diseases, have no constraints on treatment duration and are often prescribed for life. Similarly to other medical treatments, it is plausible that some patients would require FIM treatments for relatively brief periods, others may require them indefinitely, and others may cycle on and off treatments.

Most FIM treatments incorporate nutrition and culinary education. Anecdotal experiences support their value, but formal evaluation of their benefits or optimal delivery mechanisms remains limited. Other process questions are common. FIM services can vary in local context, scope, design, and methodology depending on program goals, time, resources, capacity, and partners.[23] This creates challenges for understanding which factors may or may not influence effectiveness.[94] Identifying a set of common metrics will help for comparing and pooling results across programs, together with a rigorous iterative evaluation framework to assess and build on lessons learned and strengthen future FIM programming and evaluation (**Figure 6**).

Many of these questions are not unique to FIM. For example, many widely prescribed medications were tested and approved based on studies evaluating specific populations, doses, durations, and outcomes. Following approval and clinical use, numerous

questions often remain around optimal patient eligibility, dose, alternative treatments, cost-effectiveness, impacts on health equity, and approaches to optimizing provider education and patient compliance. For example, a recent analysis found that only 32% of patients prescribed glucagon-like peptide 1 agonists for weight loss were still taking the drug at 1 year, with net 1-year health care costs among patients prescribed the drug increasing by $7,700 among noncompliant patients and by $13,200 among compliant patients (ie, indicating that the drug cost is not offset by health care savings).[95]

In the face of these realities, shared decision making is crucial to openly discuss available evidence and ambiguities, provider experience, and patient values and preferences around any treatment.[96] Comprehensive evidence reviews and recommendations, such as the American Heart Association/American College of Cardiology series of clinical practice guidelines, are also important to provide consensus statements on what is known, what remains uncertain, and how to translate this to care. Inclusion of FIM interventions in both shared decision making and future relevant practice guidelines will help clinicians, patients, payers, and policy makers understand and navigate the balance between current evidence and remaining uncertainties.

CONCLUSIONS

The rapid growth in understanding, implementation, and evaluation of FIM interventions has been remarkable, with programs and policy changes now underway across the country. The United States leads globally in this arena, with FIM projects only starting to be piloted in other nations.[97] In the face of large health, economic, and equity impacts of poor nutrition and food insecurity, FIM programs represent a compelling health care innovation to help address these burdens. Renewed attention on MNE in undergraduate, graduate, and continuing medical education is occurring and will be relevant for success and scaling. Recent national polling demonstrates widespread interest among the American public for greater attention to diet-related diseases, effective MNE for providers, and coverage of FIM services in health care (Ronit Ridberg, Tufts University, personal communication, July 27, 2023). The increasing use and costs of new weight loss drugs are also accelerating FIM: More than one-half of U.S. health plan leaders surveyed are now experimenting with FIM to minimize need for glucagon-like peptide 1 agonists.[98] Much of the initial programmatic focus and evidence has been on cardiometabolic health, making it important for clinicians in this area to understand the current landscape and coming directions. Experiences to date are promising, but current FIM programming reflects a patchwork of approaches, geographies, and payer coverage. Broader integration of FIM into health care to reach patients in need will require additional multiparty collaboration—among researchers, payers, clinicians, health care systems, patient advocates, community-based organizations, the private sector, research funders, and policy makers—to assess, optimize, and scale the interventions with greatest effects on nutrition, health, health disparities, and corresponding economic and societal burdens.

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